DS 5983: Large Language Models (Spring 2025) Roi Yehoshua
Student name: (Due) March 24, 2025

PA4: Hugging Face Transformers

In this assignment, you will gain hands-on experience with the Hugging Face Transformers library by solving a text summarization task. Text summarization is one of the most challenging NLP tasks as it requires a range of abilities, such as understanding long passages and generating coherent text that captures the main points in a document.

In this assignment, you will compare several pre-trained models from the Hugging Face model hub on text summarization and then fine-tune one of them on a custom dataset. The dataset you will use for this task is the SAMSum corpus developed by Samsung, which contains about 16,000 messenger-like dialogues and their corresponding summaries.

1. Read the Hugging Face summarization task guide.
2. Exploratory Data Analysis:
   (a) Conduct an initial exploration of the SAMSum dataset to gain insights into the characteristics of the dialogues and summaries.
   (b) Plot the length distribution of dialogues and summaries in the training set.
   (c) Display the 20 most common words in the dialogues and their frequencies.
3. Inference with Pre-trained Models:
   (a) Choose at least three pre-trained summarization models from the Hugging Face model hub, such as `facebook/bart-large-cnn` or `t5-large`.
   (b) Use the models to generate summaries for a few randomly selected dialogues. Analyze the quality of these summaries: Are they coherent? Do they capture the essential points in the conversation?
4. Define a Baseline:
   (a) Define a baseline that simply takes the first three sentences of the dialogue, often called the lead-3 baseline. You can use the `sent_tokenize` function in NLTK to split the text into sentences.
   (b) Compare the performance of the lead-3 baseline with the pre-trained models on the same dialogues using the ROUGE score.
5. Fine-Tuning:
   (a) Choose one of the pre-trained summarization models from the previous part.
   (b) Preprocess the dataset to fit it to the input format required by the chosen model, which may include tokenizing the dialogues and their summaries.
   (c) Train the model on the SAMSum corpus, monitoring its performance on the validation set, and adjust the hyperparameters as needed.

(d) Employ appropriate metrics for evaluating summarization quality, such as ROUGE scores, which compare the generated summaries against the reference summaries.

6. Evaluation and Analysis:

    (a) Evaluate the fine-tuned model's performance on the test set both quantitatively (using ROUGE score) and qualitatively (by inspecting the generated summaries).

    (b) Compare these results to the model's pre-fine-tuning performance using the same dialogues to assess the impact of the fine-tuning.

7. Write a short report covering the model choice, dataset preparation, fine-tuning process, and an analysis of the summarization performance before and after fine-tuning.